

Math 420/620, Mathematical Modeling
Homework # 2

A manufacturer of tablet computers currently sells 10,000 units per month of a basic model. The cost of manufacture is \$700/unit and the wholesale price is \$950. During the last quarter the manufacturer lowered the price \$100 in a few test markets, and the result was a 50% increase in sales—we will call this the price elasticity. The company has been advertising its products nationwide at a cost of \$50,000 per month. The advertising agency claims that increasing the advertising budget by \$5,000/month would result in a sales increase of 100 units/month. Management has agreed to consider an increase in the advertising budget to no more than \$75,000/month.

- a) Determine the price and the advertising budget that will maximize profit. Use the five-step method. Model as a constrained optimization problem. Include both a 3d plot and a contour plot of the profit function. In your search for the location of the optimal profit, be certain to
- examine the interior of the feasible region and
 - examine the boundary of the feasible region using the Lagrange Multiplier Method when and where appropriate. If you dismiss part of the boundary from further consideration without application of the Lagrange Multiplier Method, explain why.
- b) Determine the sensitivity of the decision variables (price and advertising) to price elasticity. Also interpret the sensitivities.
- c) Determine the sensitivity of the decision variables to the advertising agency's estimate of 100 new sales each time the advertising budget is increased by \$5,000/month. Also interpret the sensitivities.
- d) What is the value of the multiplier found in part (a)? What is the real world significance of the multiplier? How could you use this information to convince top management to lift the ceiling on advertising expenditures?

Notes:

$$s = 10000 + \frac{5000}{100}(950 - p) + \frac{?}{?}(a - 50000)$$

where the factor $\frac{5000}{100}$ is called the price elasticity and where we assume the following notation

- p = price (\$/computer),
- s = sales (computers/month),
- a = advertising budget (\$/month).